SUPPLEMENTARY MATERIAL

Trace-Aware Power System Scheduling with Heterogeneous Power Sources: A Learning-Based Framework

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This document accompanies the paper cited above. It provides the test-system settings, neural-network training and MILP reformulation details, and supplementary figures used to support the numerical studies.

S1 Modified IEEE 14-Bus Test System

The small-system study is based on the MATPOWER `case14` data, with branch 4–5 removed. The wind source, scheduling settings, and source-share requirement are introduced as summarized below.

Table S1: Principal system and scheduling settings.

Parameter	Value
Base power	100 MVA
Buses / branches	14 / 19
Thermal units	5
Thermal-unit buses	1, 2, 3, 6, 8
Wind farm	100 MW at Bus 3
Requirement bus	Bus 4
Horizon / interval	24 h / 1 h
Minimum output	$0.2P^{\max}$
Ramp-up/down limits	$0.6P^{\max}$/h
Minimum up/down time	3 h / 3 h
Startup cost	$3000/start
Cost approximation	Four segments

Table S2: Thermal-unit parameters.

Unit	Bus	P (MW)		c_2	c_1	c_0
		$P^{\max}$	$P^{\min}$			
TU1	1	332.4	66.48	0.0430	19	0
TU2	2	140.0	28.00	0.2500	20	0
TU3	3	100.0	20.00	0.0100	40	0
TU4	6	100.0	20.00	0.0100	41	0
TU5	8	100.0	20.00	0.0100	42	0

Table S3: Neural-network settings.

Parameter	Value
Retained samples	34,427
Train / validation / test	70% / 15% / 15%
Inputs	Total wind; TU1–TU5 outputs
Outputs	Shares at Buses 3 and 4
Hidden layers	[24, 12]
Activation	ReLU

The base active-load vector is

$$P^{\mathrm{L}}_{\mathrm{base}} = [0, 21.7, 94.2, 47.8, 7.6, 11.2, 0, 0, 29.5, 9.0, 3.5, 6.1, 13.5, 14.9]\ \mathrm{MW}. \tag{S1}$$

Table S4 gives the hourly factors used in the 24-h scheduling study.

Table S4: Hourly load and wind factors.

Hour	Load	Wind	Hour	Load	Wind
1	0.7389	0.6602	13	0.8923	0.6053
2	0.7187	0.6830	14	0.8877	0.6038
3	0.7035	0.7083	15	0.8840	0.6068
4	0.6983	0.6925	16	0.8849	0.9050
5	0.7040	0.6845	17	0.8941	0.8218
6	0.7251	0.7142	18	0.9345	0.8220
7	0.7764	0.7235	19	0.9899	0.6870
8	0.8461	0.5415	20	0.9841	0.6559
9	0.8951	0.7793	21	0.9691	0.6720
10	0.9094	0.7860	22	0.9436	0.6934
11	0.9078	0.7215	23	0.9094	0.7058
12	0.9071	0.6585	24	0.8674	0.6964

S2 Neural-Network Training and MILP Reformulation

For sample m, let $\mathbf{x}^{(m)}$ and $\mathbf{T}^{\mathrm{G},(m)}$ denote the input and tracing-label vectors. The input and output statistics are denoted by $(\boldsymbol{\mu}_x, \boldsymbol{\sigma}_x)$ and $(\boldsymbol{\mu}_y, \boldsymbol{\sigma}_y)$, respectively. The function $\sigma(\cdot)$ is the ReLU activation, Θ collects the trainable weights and biases, and $\oslash$ and $\odot$ denote elementwise division and multiplication. Training and output recovery are written as

$$\bar{\mathbf{x}}^{(m)} = (\mathbf{x}^{(m)} - \boldsymbol{\mu}_x) \oslash \boldsymbol{\sigma}_x, \qquad \bar{\mathbf{y}}^{(m)} = (\mathbf{T}^{\mathrm{G},(m)} - \boldsymbol{\mu}_y) \oslash \boldsymbol{\sigma}_y, \tag{S2a}$$

$$\mathbf{z}^{1,(m)} = \sigma(\mathbf{W}^1 \bar{\mathbf{x}}^{(m)} + \mathbf{b}^1), \qquad \mathbf{z}^{p,(m)} = \sigma(\mathbf{W}^p \mathbf{z}^{p-1,(m)} + \mathbf{b}^p), \quad p = 2, \ldots, L, \tag{S2b}$$

$$\hat{\mathbf{y}}^{(m)} = \mathbf{W}^{L+1} \mathbf{z}^{L,(m)} + \mathbf{b}^{L+1}, \qquad \Theta = \{\mathbf{W}^p, \mathbf{b}^p\}_{p=1}^{L+1}, \tag{S2c}$$

$$\min_{\Theta} \frac{1}{N_{\mathrm{sam}}} \sum_{m=1}^{N_{\mathrm{sam}}} \|\hat{\mathbf{y}}^{(m)} - \bar{\mathbf{y}}^{(m)}\|_2^2, \tag{S2d}$$

$$\widetilde{\mathbf{T}}^{\mathrm{G},(m)} = \boldsymbol{\mu}_y + \boldsymbol{\sigma}_y \odot \hat{\mathbf{y}}^{(m)}. \tag{S2e}$$

In the scheduling model, the same forward mapping is evaluated period by period by replacing the sample index m with the scheduling index t.

For a hidden neuron with affine input $\hat{z}$, ReLU output $z = \max\{\hat{z}, 0\}$, and valid bounds $\underline{h} \le \hat{z} \le \bar{h}$, introduce a binary activation variable a. Its exact mixed-integer linear representation is

$$z \ge 0, \qquad z \ge \hat{z}, \tag{S3a}$$

$$z \le \hat{z} - \underline{h}(1-a), \qquad z \le \bar{h}a, \qquad a \in \{0, 1\}. \tag{S3b}$$

When $a = 0$, the constraints give $z = 0$ and require $\hat{z} \le 0$; when $a = 1$, they give $z = \hat{z} \ge 0$. Applying these constraints neuron by neuron, followed by the affine output layer and denormalization, yields the MILP embedding used in the paper.

S3 Practical-Scale Transmission-System Case

Table S5: Principal practical-scale case settings.

Parameter	Value
Base power	100 MVA
Buses / branches	57 / 78
Thermal units	50
Wind farms	8 × 1500 MW
Wind-farm buses	7, 13, 23, 25, 42, 47, 49, 53
Total wind capacity	12,000 MW
Requirement / slack bus	55 / 38
Horizon / interval	24 h / 1 h
Line-rating multiplier	3
Cost approximation	Four segments

Table S6: Thermal-unit parameters by bus.

Bus	Units	$P^{\max}$ (MW or MW/h)	$P^{\min}$	RU/RD	c_1
3	2	1200	240	720	38.5000
8	2	1320	264	792	41.5000
9	4	1200	240	720	20.0000
11	6	4400	880	2640	30.9091
14	2	1200	240	720	40.0000
24	4	2020	404	1212	33.0693
30	4	1920	384	1152	33.7500
32	2	700	140	420	20.0000
35	2	780	156	468	20.0000
37	2	2000	400	1200	20.0000
38	6	2400	480	1440	30.0000
40	4	1900	380	1140	32.6316
46	2	2000	400	1200	20.0000
50	2	1200	240	720	40.0000
54	2	1320	264	792	40.0000
57	4	1800	360	1080	33.3333

The listed capacity and ramping values are totals over the thermal units connected to each bus. The common quadratic and constant cost coefficients are $c_2 = 0.01$ and $c_0 = 0$, respectively. The quadratic costs are represented by the four-segment linear approximation reported in Table S5.

S4 Supplementary Figures

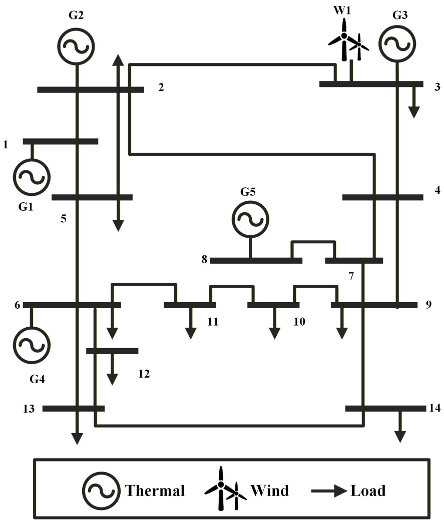

Figure S1: Topology of the modified IEEE 14-bus test system.

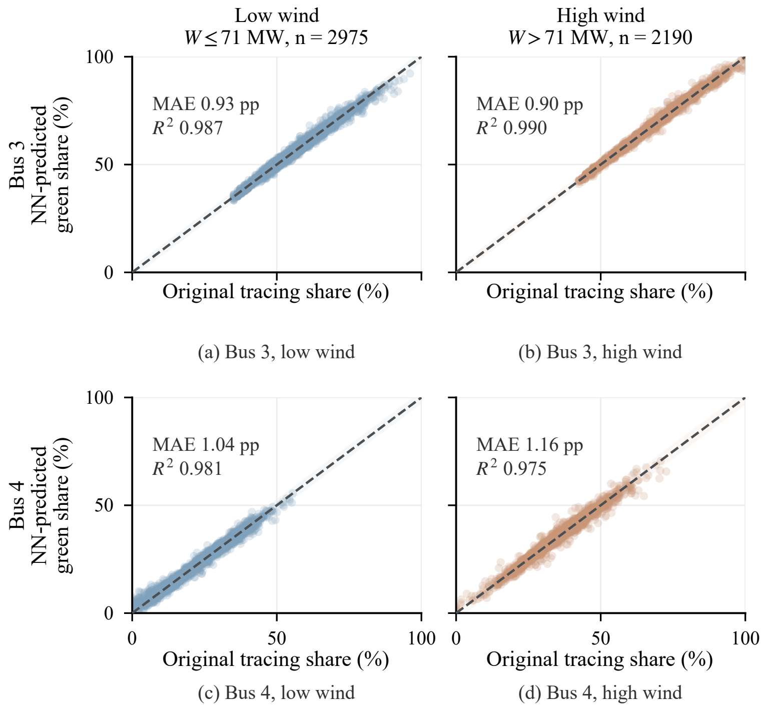

Figure S2: Held-out parity comparison under low- and high-wind conditions.